Applied Econometrics: Session 8

Synthetic Control

PhD Intensive Course

Lecture 8

April 29, 2026

How These Slides Are Structured

- The lecture is about one/few treated units observed over time.
- We build the missing counterfactual as a weighted average of untreated units.
- Each concept appears in words, a formula, and an applied interpretation.
- Technical derivations and inference details are in the appendix.

Roadmap for Today

1. Motivation
2. Setup
3. Constructing the Synthetic Unit
4. Inference and Placebos
5. Interpretation and Workflow

Why Synthetic Control?

- Many important policies affect one state, one city, one school, or one firm.
- We may observe many years, but only one treated unit.
- A single control unit is rarely credible enough.
- Synthetic control constructs a comparison unit from many untreated units.

Core idea

Instead of searching for the perfect untreated unit, build one from a weighted average of imperfect untreated units.

Running Example

Imagine a city introduces a congestion charge in year $T_0 + 1$.

- Treated unit: the city that adopts the policy.
- Outcome: traffic delays, pollution, retail sales, or public transit use.
- Donor pool: similar cities that did not adopt the policy.
- Goal: estimate what would have happened in the treated city without the policy.

Why not just compare to one city?

The closest city may match population but not income, match income but not trends, or match trends but not baseline levels.

The DiD Problem with One Treated Unit

Difference-in-differences usually asks for a treated group and a control group.

With city-level data:

$$\widehat{\tau}_{DiD} = (Y_{1,post} - Y_{1,pre}) - (Y_{0,post} - Y_{0,pre}).$$

- The formula is simple, but the control unit matters enormously.
- With one treated and one control city, we cannot average away idiosyncratic shocks.
- Standard regression standard errors can be meaningless when the effective number of units is tiny.

From One Control to a Synthetic Control

Instead of comparing city 1 to city 2, compare city 1 to

$$\text{synthetic city} = 0.35 \times \text{city A} + 0.25 \times \text{city B} + 0.40 \times \text{city C}.$$

- The weights are chosen using only pre-treatment information.
- A good synthetic city should track the treated city before treatment.
- After treatment, the gap is interpreted as the estimated policy effect.

What Makes the Method Attractive?

- Transparent: students can see which donor units receive weight.
- Visual: the treated path and synthetic path can be plotted over time.
- Design-based: credibility comes from pre-treatment fit and donor-pool logic.
- Useful when treatment is aggregate and cannot be studied with many treated units.

Teaching warning

Synthetic control is not magic. It replaces a hidden extrapolation with a visible construction that we can inspect.

Data Structure

We observe $J+1$ units over T periods.

- Unit 1 is treated after period T_0.
- Units $2, \ldots, J+1$ are untreated donor units.
- Y_{jt} is the observed outcome for unit j at time t.

Potential outcomes:

$$Y_{jt}^N = \text{outcome without treatment}, \qquad Y_{jt}^I = \text{outcome with treatment}.$$

For the treated unit and $t > T_0$,

$$\tau_{1t} = Y_{1t}^I - Y_{1t}^N.$$

Appendix details

The Missing Counterfactual

For $t > T_0$, the treated unit's observed outcome is

$$Y_{1t} = Y_{1t}^{I}.$$

The unobserved object is

$$Y_{1t}^{N},$$

the outcome the same treated unit would have had without treatment.

Synthetic-control target

Estimate Y_{1t}^{N} using untreated units that resemble the treated unit before treatment.

Donor Pool

The donor pool is the set of units allowed to contribute to the synthetic control.

A good donor unit should:

- not receive the treatment;
- not be strongly affected by spillovers from the treated unit;
- be measured consistently over time;
- be plausibly comparable in the pre-treatment period.

Practical rule

The donor pool is a research-design choice, not a software default.

Bad Donors Contaminate the Counterfactual

Do not include donors that:

- adopt the same policy soon after the treated unit;
- respond strategically to the treated unit's policy;
- have data definitions that change at the intervention date;
- are structurally impossible comparisons.

Example

If California adopts a tobacco tax, a state with a simultaneous anti-smoking campaign is not a clean donor for the no-policy path.

Pre-Treatment Information

Synthetic control chooses weights using predictors measured before treatment. Common predictors:

- lagged outcomes $Y_{j1}, \ldots, Y_{jT_0}$;
- baseline covariates such as income, demographics, or prices;
- averages over selected pre-treatment windows;
- other variables believed to forecast the outcome.

Beginner warning

Post-treatment variables must not be used to choose weights. That would build the answer into the counterfactual.

Predictor Matrices

Let X_1 be the $k \times 1$ vector of pre-treatment predictors for the treated unit.

$$X_1 = \begin{bmatrix} \text{treated pre-period outcome average} \\ \text{treated income} \\ \text{treated baseline demographics} \\ \vdots \end{bmatrix}.$$

Let X_0 be the $k \times J$ matrix with the same predictors for donor units.

$$X_0 = \begin{bmatrix} \text{donor 1 predictor} & \cdots & \text{donor J predictor} \end{bmatrix}.$$

Weighted Counterfactual

Choose weights

$$W = (w_2, \ldots, w_{J+1})'.$$

The synthetic estimate of the untreated outcome path is

$$\widehat{Y}_{1t}^{N} = \sum_{j=2}^{J+1} w_j Y_{jt}.$$

The estimated treatment effect at time $t > T_0$ is

$$\widehat{\tau}_{1t} = Y_{1t} - \widehat{Y}_{1t}^{N}.$$

Appendix details

Convex Weights

Synthetic control usually imposes:

$$w_j \geq 0 \quad \text{for all } j, \qquad \sum_{j=2}^{J+1} w_j = 1.$$

- Nonnegative weights prevent using a donor with a negative contribution.
- Sum-to-one weights make the synthetic unit an average, not an arbitrary scale change.
- Together, these constraints keep the synthetic unit inside the donor-pool range.

Intuition

The method is safer when it interpolates among observed units rather than extrapolating beyond them.

Appendix details

The Optimization Problem

A standard weight choice solves

$$\widehat{W} = \arg\min_{W} (X_1 - X_0 W)' V (X_1 - X_0 W)$$

subject to

$$w_j \geq 0, \qquad \sum_j w_j = 1.$$

- V is a diagonal matrix that says which predictors matter more.
- With lagged outcomes only and $V = I$, this is constrained least squares on the pre-treatment path.
- The fitted synthetic path is then carried into the post-treatment period.

Appendix details

Why Not Plain OLS Weights?

If we regress the treated unit's pre-treatment outcome on donor outcomes, OLS may choose:

$$\widehat{Y}_{1t}^{OLS} = -0.8\,Y_{2t} + 1.7\,Y_{3t} + 0.4\,Y_{4t} - 0.3\,Y_{5t}.$$

- Negative weights can create impossible synthetic units.
- Weights above one can stretch donors far outside observed support.
- With many donor units, OLS can overfit the pre-period and behave wildly afterward.

Practical lesson

A perfect pre-treatment fit from unconstrained OLS can be a warning sign, not a victory.

Small Numerical Illustration

Suppose the treated unit has pre-treatment predictors:

$$X_1 = (10, 4)'.$$

Donors have

donor	income	baseline outcome
A	8	3
B	12	5
C	9	4

One synthetic control is

$$0.50A + 0.25B + 0.25C = (9.25, 3.75)'.$$

It is not exact, but it is interpretable.

Pre-Treatment Fit

The central diagnostic is:

$$Y_{1t} \approx \sum_{j=2}^{J+1} \widehat{w}_j Y_{jt} \qquad \text{for } t \leq T_0.$$

- If the synthetic unit cannot track the treated unit before treatment, the post-treatment comparison is weak.
- Pre-treatment fit should be shown graphically.
- A numeric fit measure is useful, but it cannot replace the plot.

Fit measure

$$\mathrm{RMSPE}_{pre} = \sqrt{\frac{1}{T_0} \sum_{t=1}^{T_0} (Y_{1t} - \widehat{Y}_{1t}^N)^2}.$$

Appendix details

What Counts as Good Fit?

There is no universal cutoff.

- Good fit means the synthetic path captures the level and trend of the treated path.
- Small errors in irrelevant wiggles matter less than persistent pre-treatment gaps.
- If the treated unit is outside the donor support, good fit may be impossible.

Credibility statement

We trust the post-treatment gap only to the extent that the pre-treatment construction looked like a credible untreated version of the treated unit.

Treated vs. Synthetic Path

The main plot has two lines:

$$\text{treated path } Y_{1t}, \qquad \text{synthetic path } \widehat{Y}_{1t}^{N}.$$

- Before T_0: the lines should be close.
- After T_0: the vertical distance is the estimated effect.
- The treatment date should be marked clearly.

Reading the graph

If the treated line falls below the synthetic line after a tax, the estimate says the tax reduced the outcome relative to the no-tax counterfactual.

The Effect Path

A second useful plot shows the gap:

$$\widehat{\tau}_{1t} = Y_{1t} - \widehat{Y}_{1t}^{N}.$$

- This makes dynamic effects visible.
- Effects may grow, fade, reverse, or appear with delay.
- Pre-treatment gaps should hover near zero if fit is good.

Interpretation

Synthetic control estimates a time path of treatment effects, not only one average effect.

Why Usual Standard Errors Are Awkward

With one treated unit, the relevant sample size is not the number of rows in the panel.

- Repeating the same unit over many years does not create many independent treated units.
- Regression standard errors can understate uncertainty.
- We need inference that respects the small number of aggregate units.

Alternative

Use placebo exercises: pretend each donor unit was treated and compare its fake effect with the real treated effect.

Placebo Test Logic

For each donor unit q:

1. Temporarily label q as treated.
2. Build a synthetic control for q using the remaining donors.
3. Compute the placebo gap $Y_{qt} - \widehat{Y}_{qt}^{N}$.
4. Compare placebo gaps with the treated unit's gap.

Intuition

If many untreated units show gaps as large as the treated unit, the treated gap is less impressive.

Appendix details

Graphical Placebo Inference

A common figure overlays:

- the treated unit's gap path, emphasized;
- all donor placebo gap paths, shown lightly;
- a vertical line at the treatment date;
- a horizontal zero line.

Reading the figure

The treated effect is more persuasive when it becomes unusual after treatment but was not unusual before treatment.

Placebo p-Values

For a negative effect at final period T, define

$$p = \frac{1 + \sum_{q \in \mathcal{D}} 1\{\widehat{\tau}_{qT}^{placebo} \leq \widehat{\tau}_{1T}\}}{1 + |\mathcal{D}|}.$$

- The numerator counts placebo effects at least as negative as the treated effect.
- The added 1 includes the treated unit in the randomization distribution.
- With few donors, p-values are necessarily coarse.

Appendix details

Poorly Fit Placebos

Some placebo units cannot be matched well before treatment.

- Their post-treatment gaps are noisy even when there is no treatment.
- Comparing the treated unit to badly fitted placebos can be misleading.
- Researchers often report placebo graphs both with and without high pre-period RMSPE units.

Transparent rule

If you exclude placebo units with poor pre-treatment fit, state the rule before interpreting the post-treatment gaps.

RMSPE Ratio

Another diagnostic statistic is

$$\text{ratio} = \frac{\mathrm{RMSPE}_{\textit{post}}}{\mathrm{RMSPE}_{\textit{pre}}}.$$

- Large ratios mean the gap becomes much larger after treatment.
- This adjusts for units that were already hard to fit before treatment.
- It is useful for ranking the treated unit against placebo units.

What Exactly Is Identified?

Synthetic control estimates the treated unit's missing untreated path:

$$Y_{1t}^N \approx \widehat{Y}_{1t}^N = \sum_j \widehat{w}_j Y_{jt}.$$

Therefore

$$\widehat{\tau}_{1t} = Y_{1t} - \widehat{Y}_{1t}^N$$

is an estimate of the treatment effect for the treated unit at time t.

Scope

This is not automatically the average effect for all cities, states, firms, or countries.

Substantive Interpretation

After estimating the gap, ask:

- Is the sign economically meaningful?
- Is the magnitude large relative to the baseline outcome?
- Does the timing match the policy mechanism?
- Are there other events at the treatment date that could explain the gap?

Example

A tobacco-tax effect should not only be statistically unusual; it should also appear after the tax and have a plausible size given prices and consumption behavior.

Practical Workflow

1. Define the treated unit, treatment date, outcome, and estimand.
2. Build a defensible donor pool.
3. Choose pre-treatment predictors and scale them if needed.
4. Solve for convex weights.
5. Plot treated versus synthetic paths.
6. Plot the gap path.
7. Run placebo tests and report fit diagnostics.
8. Interpret effect size, timing, and limitations.

Appendix details

Choosing Predictors

A simple starting point is to match lagged outcomes:

$$X_1 = (Y_{11}, \ldots, Y_{1T_0})', \qquad X_0 = \begin{bmatrix} Y_{21} & \cdots & Y_{J+1,1} \\ \vdots & & \vdots \\ Y_{2T_0} & \cdots & Y_{J+1,T_0} \end{bmatrix}.$$

You may add covariates if they predict the outcome and are pre-treatment.

Practical warning

If one predictor is measured in thousands and another in percentages, scaling choices can dominate the weights.

Sensitivity Analysis

Report how conclusions change when:

- excluding one influential donor at a time;
- changing the pre-treatment window;
- changing predictor sets;
- dropping donors with possible spillovers;
- using alternative outcome definitions.

Goal

We do not need every specification to be identical, but the main conclusion should not depend on one fragile choice.

Common Limitations

- Poor pre-treatment fit weakens the design.
- Spillovers violate the donor-pool logic.
- Other shocks at the treatment date can be confounded with treatment.
- Convex weights cannot match treated units outside donor support.
- Many researcher choices remain: donors, predictors, windows, and fit thresholds.

Bottom line

Synthetic control is most convincing when the design story, pre-treatment fit, and placebo evidence all point in the same direction.

Synthetic Control Compared with Earlier Methods

Method	Comparison	Key assumption
DiD	Treated group versus control group before/after	Parallel trends
Matching	Treated units versus similar controls	Selection on observables
Synthetic control	Treated unit versus weighted donor pool	Donor weights recover missing untreated path

Connection

Synthetic control borrows matching's comparison logic and DiD's time-series logic, but makes the comparison unit explicit.

Python Example in the Course Folder

The companion script is:

`lectures/code/lecture-8-synthetic-control.py`.

It:

- simulates one treated unit and several donor units;
- estimates convex synthetic-control weights with projected gradient descent;
- compares treated and synthetic outcome paths;
- runs placebo reassignments across donor units;
- reports a finite-sample placebo p-value.

Run with plots:

```
python lectures/code/lecture-8-synthetic-control.py --plot
```

What Students Should Be Able to Do

After this lecture, students should be able to:

- explain synthetic control as a missing-counterfactual estimator;
- define treated unit, donor pool, pre-treatment period, and effect path;
- write the convex-weight optimization problem;
- interpret treated-versus-synthetic and gap plots;
- explain placebo tests and pre-treatment fit diagnostics;
- name the main limitations and credibility checks.

Main Takeaways

- Synthetic control builds a counterfactual unit from untreated donor units.
- Convex weights make the construction transparent and reduce extrapolation risk.
- Pre-treatment fit is the central diagnostic.
- The effect is a post-treatment gap between the treated and synthetic paths.
- Placebo tests ask whether the treated gap is unusual relative to fake treated units.

Sources and References

- Matheus Facure Alves, *Causal Inference for the Brave and True*, Chapter 15: Synthetic Control. `https://matheusfacure.github.io/python-causality-handbook/15-Synthetic-Control.html`
- Abadie, Diamond, and Hainmueller (2010), "Synthetic Control Methods for Comparative Case Studies." *Journal of the American Statistical Association*.
- Abadie (2021), "Using Synthetic Controls: Feasibility, Data Requirements, and Methodological Aspects." *Journal of Economic Literature*.

Appendix Map

- A1: Potential-outcomes setup for synthetic control.
- A2: Why weighted donor outcomes estimate the missing path.
- A3: Convex weights and the convex hull.
- A4: Constrained least-squares optimization.
- A5: RMSPE diagnostics.
- A6: Placebo inference logic.
- A7: Practical reporting checklist.

Appendix A1: Potential-Outcomes Setup

For the treated unit 1:

$$Y_{1t} = \begin{cases} Y_{1t}^N, & t \leq T_0, \\ Y_{1t}^I, & t > T_0. \end{cases}$$

For donor units $j = 2, \ldots, J+1$:

$$Y_{jt} = Y_{jt}^N \quad \text{for all } t.$$

The post-treatment effect is

$$\tau_{1t} = Y_{1t}^I - Y_{1t}^N, \qquad t > T_0.$$

Since Y_{1t}^I is observed after treatment, identification reduces to approximating Y_{1t}^N.

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Appendix A2: Weighted Donor Counterfactual

Suppose there exists a donor-weight vector $W^\star$ such that the untreated treated-unit path is well approximated by

$$Y_{1t}^N \approx \sum_{j=2}^{J+1} w_j^\star Y_{jt}^N .$$

Since donor units are untreated,

$$Y_{jt}^N = Y_{jt}.$$

Therefore

$$Y_{1t}^N \approx \sum_{j=2}^{J+1} w_j^\star Y_{jt}.$$

Replacing $W^\star$ with the estimated pre-treatment weights gives

$$\widehat{\tau}_{1t} = Y_{1t} - \sum_{j=2}^{J+1} \widehat{w}_j Y_{jt}.$$

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Appendix A3: Convex Hull Logic

A convex combination has

$$w_j \geq 0, \qquad \sum_j w_j = 1.$$

Let Z_j be any scalar predictor. The synthetic predictor is

$$Z_{syn} = \sum_j w_j Z_j.$$

If $Z_{\min} \leq Z_j \leq Z_{\max}$ for all donors, then

$$Z_{\min} = \sum_j w_j Z_{\min} \leq \sum_j w_j Z_j \leq \sum_j w_j Z_{\max} = Z_{\max}.$$

So the synthetic unit cannot lie outside the donor range on any one predictor.

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Appendix A4: Constrained Least Squares

With $V = I$, the weight problem is

$$\min_{W} (X_1 - X_0 W)'(X_1 - X_0 W)$$

subject to

$$\iota' W = 1, \qquad W \geq 0.$$

Ignoring inequality constraints for a moment, the Lagrangian is

$$\mathcal{L}(W, \lambda) = (X_1 - X_0 W)'(X_1 - X_0 W) + \lambda(\iota' W - 1).$$

First-order conditions:

$$-2X_0'(X_1 - X_0 W) + \lambda\iota = 0, \qquad \iota' W = 1.$$

Nonnegativity adds complementary-slackness conditions and is usually handled numerically.

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Appendix A5: RMSPE Diagnostics

Pre-treatment RMSPE:

$$\mathrm{RMSPE}_{pre} = \sqrt{\frac{1}{T_0}\sum_{t=1}^{T_0}(Y_{1t} - \widehat{Y}_{1t}^{N})^2}.$$

Post-treatment RMSPE over periods $T_0 + 1, \ldots, T$:

$$\mathrm{RMSPE}_{post} = \sqrt{\frac{1}{T - T_0}\sum_{t=T_0+1}^{T}(Y_{1t} - \widehat{Y}_{1t}^{N})^2}.$$

The ratio $\mathrm{RMSPE}_{post}/\mathrm{RMSPE}_{pre}$ is large when the post-treatment gap is large relative to pre-treatment mismatch.

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Appendix A6: Placebo Inference as Randomization Logic

Let $\mathcal{D}$ be donor units. For each $q \in \mathcal{D}$, compute a placebo statistic S_q, such as final-period gap or RMSPE ratio.

If the treatment assignment were exchangeable across units, then the treated statistic S_1 should look like one draw from

$$\{S_1, S_q : q \in \mathcal{D}\}.$$

For a lower-tail effect:

$$p = \frac{1 + \sum_{q \in \mathcal{D}} 1\{S_q \leq S_1\}}{1 + |\mathcal{D}|}.$$

This is exact only under the maintained exchangeability/design assumptions.

Appendix A7: Reporting Checklist

A complete synthetic-control application should report:

- treated unit, treatment date, outcome, and analysis window;
- donor-pool inclusion and exclusion rules;
- predictor variables and scaling choices;
- estimated donor weights;
- treated-versus-synthetic path plot;
- gap plot and effect magnitudes;
- pre-treatment RMSPE and placebo evidence;
- limitations: spillovers, other shocks, poor fit, and sensitivity.

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